

Alekhya Dabbiru

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ANALYTICS | MACHINE LEARNING | BUSINESS INTELLIGENCE | PYTHON | SQL | ETL

SKILLS

- **Programming and Libraries:** Python, SQL, R, Pandas, NumPy, Scikit-learn
- **Tools and Platforms:** Git, GitHub, Docker, Jupyter Notebook
- **Machine Learning and Statistics:** Classification, Regression, Clustering, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation, Forecasting
- **Data Engineering and Cloud:** ETL Pipelines, Data Warehousing, Data Validation, Snowflake, Databricks, Spark, AWS S3, Data Transformation, Data Integration, Data Integrity
- **Analytics and Business Intelligence:** EDA, Statistical Analysis, Predictive Analytics, A/B Testing, KPI Reporting, Tableau, Power BI, Excel, Dashboard Development, Stakeholder Reporting

EXPERIENCE

Data Scientist | Tacoma Power

Jan 2026 - Jun 2026

- Engineered Python, SQL, and Snowflake ETL pipelines to consolidate **150K+ inspection records**, reducing processing time by **~30%** and improving data integrity for predictive maintenance analysis.
- Modeled equipment failure patterns using predictive analytics and anomaly review, supporting risk-based maintenance planning tied to an estimated **15% reduction in unplanned outages**.
- Automated Tableau dashboards and KPI monitoring workflows, reducing manual reporting effort by **~40%** and improving visibility into pipeline health and operational risk indicators.

Data Analyst | Manuh Global Technologies Pvt Ltd

Nov 2023 - Aug 2024

- Engineered Python and SQL validation, standardization, and preprocessing workflows across **60K+ records**, improving data quality by **~25%** for downstream modeling and analysis.
- Analyzed trends and forecasts across **7+ datasets**, reducing report turnaround time by **~20%** through automated extraction and optimized SQL queries for recurring operational planning.
- Automated Power BI and Tableau reporting workflows, increasing stakeholder adoption by **~30%** and making model inputs and KPI results easier to interpret across recurring business reviews.

Data Analyst | Edumoon, Andhra University

Nov 2022 - Oct 2023

- Analyzed engagement behavior across **7 campuses**, identifying usage and retention patterns that contributed to a **10-12% increase in active platform adoption** through behavioral segmentation and stakeholder-ready usage reporting.
- Built a centralized data pipeline that reduced manual processing and data retrieval time by **35-40%**, accelerating downstream analytics and product analysis for faster retention and feature evaluation.

PROJECTS

Parking Flow Intelligence | Python, Pandas, Scikit-learn, Random Forest, Time-Series Forecasting

- Improved occupancy forecasting from **0.874 to 0.974 R-squared** with **1.71% MAE** and **3.14% RMSE** by engineering lag features, rolling averages, and Random Forest models across **35K+ records from 30 locations**.
- Enabled congestion-risk monitoring across **30 parking locations** by building a validated ETL and forecasting pipeline that generated automated visual summaries and operational reports.

Food Delivery Cancellation and Pricing Analysis | Python, Statistical Experimentation, SciPy, XGBoost, CatBoost

- Identified the **\$20 penalty** as the strongest strategy, producing a **10.03% cancellation rate** and **\$14.05 average profit per order** by testing three fee groups with z-tests and profit analysis.
- Reached **~0.78 precision** for high-risk cancellation targeting by engineering features and tuning Random Forest, XGBoost, and CatBoost models.

Airline Delay Control Tower ETL | Python, SQL, SQLite, ETL, Operational Analytics

- Prepared **500,000 airline records** for repeatable analytics by building an ETL pipeline with validation, transformation, dimensional modeling, and SQL-ready loading.
- Prioritized recovery across **2,777 routes** by combining delay frequency, cancellation risk, and traffic volume into route-risk and performance reports.

AI Incident Risk Analyzer | Python, NLP, TF-IDF, K-Means, Text Analytics

- Categorized **1,465 real-world AI incidents** across safety, privacy, governance, bias, and misinformation risks by applying text preprocessing and TF-IDF vectorization.
- Flagged **145 high-severity incidents** and **8 risk clusters** by combining severity scoring, K-Means, Truncated SVD, and cosine-similarity search.

CERTIFICATIONS

- **Microsoft Azure AI Fundamentals**
- **Generative AI Fundamentals** - Databricks
- **Artificial Intelligence A-Z 2025 and Machine Learning A-Z** - Udemy

EDUCATION

Master of Science in Data Science | Seattle University, Seattle, WA

Sept 2024 - Jun 2026

Relevant Coursework: Statistical Modeling, Machine Learning, Database Management Systems, Big Data, Deep Learning

Bachelor of Engineering in Computer Science | Andhra University, Visakhapatnam, India

Jul 2020 - May 2024